

Extended RNA-seq analysis and candidate-gene prioritization

Methodological report - Candida parapsilosis hypoxia response

This report documents the two extended R Markdown workflows used after the ENS practical. Its purpose is to describe what was computed, the methodological rationale for each additional step, the main parameter choices, and how the resulting evidence was integrated for candidate prioritization. It is a methods report rather than a presentation summary.

1. Scope of the reanalysis and data flow

A key distinction is that the raw-read practical and the differential-expression analysis do not start from the same type of input. The extended scripts do not reconstruct the complete experiment from the small FASTQ subset used to learn alignment/counting. Both extended R workflows start from the supplied RNA-seq count matrix and its design table.

- RNA-seq input: count_data_diffAnalysis_no0counts.txt.
- Metadata input: design.txt.
- The rendered extended QC report contains 5,788 genes and 10 RNA-seq samples: 6 normoxic (N) and 4 hypoxic (H).
- The contrast is explicitly oriented as H versus N; positive log2 fold changes therefore indicate higher expression in hypoxia.
- A separate microarray matrix of replicate log(H/N) ratios is analyzed with limma and later integrated with RNA-seq.

The overall strategy is deliberately two-stage. Script 01 asks whether the supplied count-level data are sufficiently coherent for interpretation. Script 02 then challenges the differential-expression results using alternative effect estimates, sample perturbation, an independent RNA-seq method, cross-platform evidence, and several multi-criteria ranking strategies.

Stage	Input	Main question	Role in selection
Script 01	RNA-seq counts + design	Can the sample-level data and candidate signals be trusted?	Diagnostic QC; generally does not redefine the main gene list.
Script 02: DE	Same RNA-seq counts	Are effect estimates significant, large, precise and stable?	Creates additional evidence for each gene.
Script 02: validation	edgeR + microarray	Do conclusions survive another model and another technology?	Measures methodological and cross-platform reproducibility.
Script 02: ranking	Integrated evidence table	Which genes remain convincing under multiple criteria?	Produces rank consensus, percentile score, Pareto set and weight sensitivity.

2. Script 01 - extended RNA-seq preprocessing/QC

Despite the filename 'preprocessing', this script is best described as count-level normalization and quality control. It does not perform FASTQ trimming, alignment, or read counting. Most additions are diagnostic: their purpose is to detect reasons not to trust the downstream analysis.

2.1 Input validation and factor orientation

Method. The count matrix and design table are loaded, sample names are matched exactly, and the metadata are reordered to the count-matrix column order.

Why it was added. A silent mismatch between count columns and phenotype labels would invalidate every downstream model. Making normoxia the explicit reference also prevents ambiguity in fold-change orientation.

Implementation in the script. The script uses `stopifnot()` to verify sample matching, then relevels condition with N as the reference. It prints matrix dimensions, condition counts, and the number of zero entries.

How to interpret it. After releveling, the intended biological comparison is H/N.

Limitation / caution. This is a reproducibility/safety improvement, not a statistical test.

2.2 Raw library-size inspection

Method. Total mapped counts are summed for each sample and displayed as a bar plot.

Why it was added. Sequencing depth can differ substantially among libraries. Large differences are not automatically errors, but they motivate normalization and can reveal unusual samples.

Implementation in the script. `library_size = colSums(countData)`; the bar plot is grouped by condition.

How to interpret it. The plot should be interpreted together with size factors and downstream sample QC, not used alone to delete libraries.

2.3 DESeq2 normalization and model fitting

Method. A `DESeqDataSet` is built with `design ~ condition` and processed with the canonical `DESeq()` pipeline.

Why it was added. DESeq2 estimates size factors, gene-wise/dispersion trends and the negative-binomial model required for differential-expression inference.

Implementation in the script. Counts are rounded and passed to `DESeqDataSetFromMatrix()`; `DESeq(dds)` replaces the more fragmented manual `sequence estimateSizeFactors -> estimateDispersions -> nbinomWaldTest`.

How to interpret it. Normalized counts correct for library-size/composition effects; they are not TPM/FPKM and should not be interpreted as absolute expression.

2.4 Distribution before and after normalization

Method. Per-sample distributions of $\log_2(\text{count}+1)$ are compared before and after DESeq2 normalization.

Why it was added. This visualizes whether gross library-scale differences are reduced by normalization and avoids the original script's hard-coded assumption of 5,788 genes/sample ordering.

Implementation in the script. A helper function constructs the long-format table using metadata lookup by sample name rather than substrings or fixed row ranges.

How to interpret it. More comparable distributions after normalization are reassuring, but identical distributions are neither expected nor required.

2.5 Dispersion diagnostics

Method. DESeq2's fitted dispersion estimates are visualized with `plotDispEsts()`.

Why it was added. RNA-seq biological variability is modeled through the negative-binomial dispersion. Abnormal dispersion behavior can indicate model/data problems.

Implementation in the script. The plot shows gene-wise estimates, fitted trend and final/shrunken dispersion estimates.

How to interpret it. This is a global diagnostic; a gene can still be uncertain even when the overall dispersion fit looks reasonable.

2.6 Variance-stabilizing transformation and PCA

Method. Counts are variance-stabilized and PCA is performed on the blind transformation for exploratory sample QC.

Why it was added. Raw count variance grows with the mean. VST makes sample-to-sample structure easier to visualize. `blind=TRUE` prevents the experimental design from influencing the exploratory transformation.

Implementation in the script. `vst(dds, blind=TRUE)`; `plotPCA(..., returnData=TRUE)` is redrawn with sample labels and the percentage of variance explained by PC1 and PC2. A `blind=FALSE` VST is also created for downstream candidate visualization.

How to interpret it. Replicates should generally show coherent structure. PCA is not a formal outlier test and separation by condition is not a prerequisite for valid DE.

2.7 Pearson and Spearman replicate correlations

Method. Pairwise sample correlations are computed on the blind VST matrix using both Pearson and Spearman coefficients.

Why it was added. Pearson tests similarity of transformed values under a linear relationship, whereas Spearman tests similarity of ranks. Agreement between them makes the sample relationships less dependent on one correlation definition.

Implementation in the script. `cor(vmat, method='pearson')` and `cor(vmat, method='spearman')`, displayed as heatmaps.

How to interpret it. Unexpectedly low within-condition correlations can flag a sample for investigation, but correlation alone should not justify removal.

2.8 Sample-to-sample distances

Method. Euclidean distances are calculated between VST-transformed sample profiles.

Why it was added. Distance heatmaps provide a complementary view to correlation and PCA and can make an isolated sample visually obvious.

Implementation in the script. `dist(t(vmat))` followed by a heatmap.

How to interpret it. This is descriptive; it does not assign an automatic outlier threshold.

2.9 Cook's-distance diagnostics

Method. For each gene, the maximum DESeq2 Cook's distance across samples is calculated.

Why it was added. Cook's distance measures the influence of individual observations on a fitted model. A large value warns that a gene-level result may be disproportionately affected by one sample.

Implementation in the script. `cooks <- assays(dds)[["cooks"]]`; `maxCook <- apply(cooks, 1, max, na.rm=TRUE)`. The distribution is shown on a log10 x-axis and the largest values are listed.

How to interpret it. A high Cook's distance is a warning for inspection, not proof that the gene or sample should be discarded.

2.10 Candidate-level normalized-count plots

Method. Each candidate is plotted as normalized counts by condition, with a boxplot plus individual jittered samples on a log scale.

Why it was added. A single summary statistic can hide whether a fold change is consistently supported by biological replicates. Showing each point exposes heterogeneity and potential single-sample effects.

Implementation in the script. DESeq2 `plotCounts(..., returnData=TRUE)` supplies normalized counts; ggplot displays individual samples.

How to interpret it. These plots are descriptive and should be interpreted alongside model-based uncertainty.

2.11 Low-count filtering sensitivity

Method. The analysis is repeated after retaining genes with at least 10 counts in at least 3 samples.

Why it was added. Very weakly expressed genes contain limited information and can produce unstable estimates. The purpose is to test whether the main fold-change conclusions are sensitive to a reasonable prefilter.

Implementation in the script. `keep_10x3 <- rowSums(countData >= 10) >= 3`; a new DESeq2 model is fitted and unfiltered versus filtered log2FC values are plotted.

How to interpret it. Strong agreement indicates that conclusions are not strongly dependent on weakly informative genes.

Limitation / caution. The script explicitly treats this as a sensitivity analysis; it does not replace the main analysis.

3. Script 02 - extended differential-expression analysis

Script 02 uses the same supplied RNA-seq count matrix but adds methods designed to answer progressively harder questions: is the effect real, is it large enough to matter, is it precise, is it stable to removal of a replicate, does another RNA-seq framework agree, and does an independent microarray platform support the same direction and magnitude?

3.1 Main DESeq2 contrast and diagnostic variables

Method. DESeq2 is fitted with `design ~ condition` and results are extracted for H versus N.

Why it was added. This preserves the practical's core analysis while making the orientation and downstream variables explicit.

Implementation in the script. `alpha_main=0.01` and `lfc_visual=2` are retained for comparison with the practical. The script stores `direction`, `-log10(padj)`, and a simple precision proxy `|LFC|/SE`.

How to interpret it. `padj` controls multiple testing; `log2FC` measures effect magnitude; `lfcSE` measures uncertainty. None alone is sufficient for prioritization.

3.2 P-value, volcano and MA diagnostics

Method. The raw p-value distribution, volcano plot and MA plot are regenerated using named columns and explicit cutoffs.

Why it was added. These provide standard diagnostics and preserve continuity with the practical while avoiding brittle column-number indexing.

Implementation in the script. `EnhancedVolcano` is used when installed; otherwise ggplot is used. The MA plot uses `alpha_main` and a fixed y range.

How to interpret it. These plots visualize the DE landscape; they are not the new ranking method.

3.3 Log2-fold-change shrinkage

Method. Raw DESeq2 log2FC estimates are replaced for ranking/interpretation by shrunken estimates. `apeglm` is preferred; DESeq2's normal shrinkage is the fallback.

Why it was added. Large raw effects from low-information or noisy genes can be exaggerated. Shrinkage pulls uncertain effects more strongly toward zero while preserving well-supported large effects.

Implementation in the script. `lfcShrink(dds, coef='condition_H_vs_N', type='apeglm')` when `apeglm` is installed. Raw versus shrunken effects are plotted.

How to interpret it. A gene whose LFC changes little after shrinkage has an effect supported by the data; strong shrinkage signals substantial uncertainty.

Limitation / caution. Shrinkage stabilizes effect estimates; it is not simply another p-value filter.

3.4 Formal effect-size testing

Method. Instead of only testing $H_0: \text{LFC}=0$ and then applying a hard fold-change cutoff, `DESeq2` directly tests whether $|\text{LFC}|$ exceeds 1.

Why it was added. Statistical significance can occur for tiny effects. The test asks whether the magnitude is larger than a biologically moderate threshold (>2 -fold).

Implementation in the script. `results(..., lfcThreshold=1, altHypothesis='greaterAbs', alpha=0.05)`. In the rendered run, 134 genes had evidence for $|\log_2\text{FC}|>1$ at $\text{FDR}<0.05$.

How to interpret it. This separates 'detectably non-zero' from 'supported to be meaningfully large'.

Limitation / caution. The threshold of 1 is a methodological choice, not a universal biological law.

3.5 Optional `apeglm` s-values

Method. When `apeglm` is available, s-values are computed with the same LFC threshold.

Why it was added. The s-value summarizes false-sign-or-small risk: it is useful when the goal is to control errors in effect direction and whether the effect exceeds a chosen threshold.

Implementation in the script. `lfcShrink(..., lfcThreshold=1, svalue=TRUE)`.

How to interpret it. Lower values support a stable sign and sufficiently large effect.

Limitation / caution. This is supplementary evidence and was not the sole basis of candidate selection.

3.6 Abundance, uncertainty and Cook's distance

Method. The script relates shrunken effect magnitude to `baseMean` and `lfcSE` and attaches maximum Cook's distance to every gene.

Why it was added. This makes it possible to distinguish spectacular but poorly measured effects from large effects supported by abundant, precise data.

Implementation in the script. Scatter plots show $\log_{10}(\text{baseMean}+1)$ versus $|\text{shrunkenLFC}|$ and $|\text{shrunkenLFC}|$ versus `lfcSE`; `maxCook` is merged into the results table.

How to interpret it. High abundance is not automatically biological importance; it is one information/precision dimension.

3.7 Threshold-sensitivity grid

Method. The number of selected genes is recalculated over FDR values 0.001, 0.005, 0.01, 0.05, 0.10 and $|\text{shrunkenLFC}|$ cutoffs 0, 0.5, 1, 1.5, 2 and 2.5.

Why it was added. Hard cutoffs create artificial cliffs: a gene at $\text{LFC}=1.99$ is almost indistinguishable biologically from one at 2.01. Sensitivity analysis shows whether conclusions depend on a single arbitrary boundary.

Implementation in the script. An `expand.grid()` of FDR and LFC thresholds is evaluated globally and separately for candidate genes, producing heatmaps of gene counts and candidate stability.

How to interpret it. Candidates selected across many reasonable threshold combinations are less threshold-dependent.

3.8 Leave-one-sample-out robustness

Method. DESeq2 is refitted once for every omitted RNA-seq sample. With 10 samples, this produces 10 refits, each using the remaining 9.

Why it was added. This directly tests whether a candidate's differential-expression conclusion depends on any one biological replicate.

Implementation in the script. For each omitted sample the script stores log2FC and absolute-effect rank. It then summarizes median LFC, LFC SD, min/max LFC, proportion of refits with $\text{padj} < 0.05$, median rank and worst rank.

How to interpret it. `loo_propFDR05` near 1 and a small `loo_sdLFC` indicate a signal that survives sample perturbation.

Limitation / caution. LOO is a robustness analysis, not a substitute for collecting additional independent biological replicates.

3.9 Independent RNA-seq analysis with edgeR

Method. The same raw count matrix is reanalyzed with edgeR's quasi-likelihood workflow.

Why it was added. If a candidate remains strong under another established count-model implementation, the conclusion is less dependent on DESeq2-specific estimation choices.

Implementation in the script. `DGEList -> filterByExpr -> TMM normalization -> model.matrix(~group) -> estimateDisp -> glmQLFit -> glmQLFTest`. The logFC orientation is explicitly corrected to H/N if necessary.

How to interpret it. Agreement of effect direction/magnitude and low edgeR FDR constitute methodological reproducibility.

Limitation / caution. DESeq2 and edgeR are not independent experiments; they analyze the same samples and both use negative-binomial models.

4. Microarray reanalysis and cross-platform integration

4.1 Standard, robust and effect-size-aware limma

Method. The supplied microarray matrix contains replicate log(H/N) ratios. An intercept-only limma model tests whether the mean ratio differs from zero.

Why it was added. The microarray provides a technologically distinct measurement of the same biological contrast and can therefore expose platform-specific disagreement.

Implementation in the script. Three variants are fitted: standard `eBayes()`, robust `eBayes(robust=TRUE)`, and `limma treat(lfc=1)`. Replicate mean, SD, minimum and maximum are also calculated.

How to interpret it. Standard limma reproduces the practical logic; robust limma tests sensitivity to outlying variances; `TREAT` asks whether the effect exceeds a chosen magnitude.

Limitation / caution. Cross-platform agreement strengthens confidence but does not prove biological causality.

4.2 Cross-platform agreement metrics

Method. RNA-seq shrunken LFC and microarray logFC are merged by gene. Directional agreement and absolute platform difference are calculated.

Why it was added. A candidate measured similarly by two technologies is less likely to be an artifact of a platform-specific bias.

Implementation in the script. `direction_same = sign(shrunkLFC)==sign(array_logFC)`; `platform_difference = |shrunkLFC-array_logFC|`. Pearson and Spearman correlations, a scatter plot, and a Bland-Altman-style plot are produced.

How to interpret it. Pearson assesses linear agreement; Spearman assesses rank agreement; `platform_difference` is used later as a gene-level reproducibility criterion.

Limitation / caution. Microarray and RNA-seq have different dynamic ranges and technical biases, so exact numerical equality is not expected.

4.3 Discordant, rescued and fragile genes

Method. The script explicitly searches for evidence against naive selection as well as evidence in favor of candidates.

Why it was added. A robust prioritization should identify genes that fail because of cross-platform disagreement or uncertainty, and genes that were missed only because of the practical's hard thresholds.

Implementation in the script. Discordant genes require $|\text{effect}| \geq 1$ on both platforms but opposite directions. 'Rescued' genes fail the practical filter yet have RNA-seq $\text{padj} < 0.05$, $|\text{shrunkLFC}| > 1$, microarray $\text{FDR} < 0.05$ and matching direction. 'Fragile' practical hits are flagged for low abundance (bottom 20%), high lfcSE (top 20%) or poor microarray support.

How to interpret it. These categories are diagnostic labels for prioritization, not formal biological classes.

5. Multi-criteria candidate ranking

The central problem is that candidate quality is multidimensional. A p-value, a fold change, a standard error, cross-platform agreement and leave-one-out stability are measured on incompatible scales. The script therefore uses several complementary integration strategies rather than pretending that one raw quantity is sufficient.

5.1 Rank-product consensus

Method. Genes are ranked separately by DESeq2 effect, DESeq2 FDR, microarray effect, microarray FDR and, when available, edgeR effect rank. The geometric mean of available ranks is used as the rank product.

Why it was added. Ranks remove the unit/scale problem: a p-value cannot be averaged sensibly with a fold change. The geometric mean favors genes that are consistently well ranked across evidence types.

Implementation in the script. `rank_high()` is used when larger is better; `rank_low()` when smaller is better. `rank_product = exp(mean(log(valid ranks)))`. At least two finite ranks are required.

How to interpret it. A small rank product indicates consistently strong relative performance.

Limitation / caution. Rank methods discard information about the absolute distance between genes.

5.2 Percentile-based statistical score

Method. Each evidence dimension is converted to a 0-1 percentile and the available scores are averaged.

Why it was added. Percentiles put heterogeneous criteria on a common relative scale while keeping the initial score purely statistical rather than embedding biological annotation.

Implementation in the script. The score includes $|\text{shrunkLFC}|$, padj , lfcSE , $\log_{10}(\text{baseMean})$, $|\text{microarray LFC}|$, microarray FDR, platform difference, LOO support and edgeR FDR. Higher percentiles always represent more desirable evidence. `stat_score` is the row mean ignoring missing values.

How to interpret it. A high `stat_score` means broadly strong evidence across available dimensions.

Limitation / caution. The equal-weight mean is itself a choice; this is why Pareto and weight-sensitivity analyses are added.

5.3 Pareto-front analysis

Method. A gene is Pareto-dominated if another gene is at least as good on every selected criterion and strictly better on at least one.

Why it was added. Pareto analysis identifies defensible trade-offs without assigning subjective coefficients to criteria.

Implementation in the script. The script uses `score_effect`, `score_significance`, `score_precision`, `score_array_significance` and `score_platform_agreement`. Only complete finite cases are used; to keep the $O(n^2)$ comparison tractable and focused, at most the 500 highest `stat_score` genes enter the Pareto analysis. The rendered run identified 52 non-dominated genes.

How to interpret it. Pareto-front membership means 'not clearly beaten on all chosen dimensions'; it does not mean 'biologically best'.

Limitation / caution. Results depend on which criteria define the frontier and on the preselection of the top 500 statistical candidates.

5.4 Decision-weight sensitivity: 5,000 coefficient sets

Method. The same five decision criteria are combined under 5,000 randomly generated positive weight vectors that each sum to one.

Why it was added. Any single weighted score embeds subjective preferences. Repeating the ranking across many plausible priorities asks whether a gene remains attractive even when another analyst values the criteria differently.

Implementation in the script. `n_weight_draws=5000` and `set.seed(2026)`. Exponential random values are generated with `rexp()`, normalized row-wise to sum to one, and multiplied by the criterion matrix. For each draw, the script records whether each gene enters the top 5 or top 10. `P_top5_weights` and `P_top10_weights` are the resulting proportions.

How to interpret it. `P_top5_weights=0.99` means the gene appears in the top five under about 99% of the sampled weighting schemes.

Limitation / caution. This is a sensitivity analysis over a chosen weight-generating distribution; it is not a posterior probability that the gene is biologically correct.

5.5 Combined robustness

Method. Sample robustness and decision robustness are combined using a geometric mean.

Why it was added. A candidate should ideally survive both perturbation of the data (leave-one-out) and perturbation of the analyst's weighting preferences.

Implementation in the script. `combined_robustness = sqrt(loo_propFDR05 * P_top5_weights)`. The final table is sorted first by this quantity and then by `stat_score`.

How to interpret it. The geometric mean penalizes a gene that is strong on only one of the two dimensions.

Limitation / caution. This is a constructed prioritization metric, not a standard inferential statistic.

6. Additional exploratory outputs

- Five-gene PCA: PCA using only the candidate panel. The script explicitly labels this exploratory because n is small; it must not be presented as a validated classifier.
- Two-gene marker search: among the 50 strongest induced and 50 strongest repressed genes, the script searches for a VST difference score that best separates H and N relative to within-group variability. This is hypothesis-generating and highly susceptible to overfitting in a small dataset.
- Evidence heatmap: displays the 0-1 percentile evidence profile for candidate genes without clustering, making it easy to see which criterion supports or weakens each candidate.

- GSEA preparation: exports all genes with finite DESeq2 Wald statistics as a ranked list. No enrichment is run automatically because *C. parapsilosis* gene-set definitions must be supplied and documented separately.

7. Optional upstream/BAM sensitivity plan

Script 02 also contains a non-executed command-line plan for testing whether read-counting choices could alter the count matrix. These commands were included as a methodological extension but are not part of the results unless they are actually run on the original BAM files.

Command/test	Question	Important caution
samtools flagstat / stats	Are mapping summaries broadly acceptable?	Diagnostic only.
RSeQC infer_experiment.py	What is the library strandedness?	Use before applying strand-specific counting.
bedtools multicov -s	Does strand-aware counting change results?	Only biologically valid if library orientation warrants it.
bedtools multicov -split	Does treatment of split alignments affect counts?	Relevant to RNA-seq exon/gene counting.
bedtools multicov -q 20 / -q 30	Are conclusions sensitive to mapping-quality thresholds?	Do not choose the threshold that favors preferred genes.
featureCounts	Does an independent counting implementation reproduce counts?	Annotation and strandedness settings must be matched/documented.

8. How the statistical workflow relates to the final five genes

The extended ranking was not intended to mechanically dictate the biological panel. The project used two complementary routes.

- Biological-first route: practical DE lists -> stricter filtering -> GO enrichment -> STRING/MCL functional clusters -> literature/annotation -> initial biologically interesting candidates.
- Evidence-first route: extended QC -> LFC shrinkage/effect-size testing -> threshold sensitivity -> leave-one-out -> edgeR -> microarray integration -> rank consensus/percentile score -> Pareto -> 5,000 weight schemes.
- Final prioritization: convergence between robust statistical evidence and a testable biological rationale, while respecting the instruction to exclude CPAR2_404850.

This separation is important methodologically. Biological annotation was deliberately kept out of the initial statistical percentile score, reducing the risk of constructing a score that merely rewards genes already considered interesting.

9. Methodological limitations

- The additional QC mostly validates the suitability of the data; it does not itself create an independent experiment.
- Leave-one-out evaluates sensitivity to existing samples but cannot replace additional biological replication.
- DESeq2-edgeR concordance is methodological reproducibility on the same RNA-seq data, not independent biological replication.
- RNA-seq-microarray agreement is stronger cross-platform evidence, but the platforms differ in measurement properties and exact numerical effects need not match.
- Effect-size thresholds (for example $|\log_2FC| > 1$) are chosen conventions and should be justified biologically.
- The percentile score gives equal weight to available dimensions before the later sensitivity analysis; missing data can mean different genes are averaged over different subsets.
- Pareto results depend on the chosen criteria and, in this implementation, the top-500 statistical preselection.

- The 5,000 random weight analysis measures robustness to a defined family of weighting schemes, not the probability that a gene is truly causal.
- Combined robustness is a custom prioritization index, not a standard test statistic.
- No transcriptomic analysis proves function. qRT-PCR, genetic perturbation/complementation, hypoxia phenotyping and pathway-specific assays are required for causal validation.

10. Reproducibility and outputs

- Random seed for weight sensitivity: `set.seed(2026)`.
- Computational switches: `run_leave_one_out=TRUE`; `run_edgeR=TRUE`; `run_weight_sensitivity=TRUE`; `n_weight_draws=5000`.
- Main practical-comparison cutoffs: `alpha_main=0.01`; `lfc_visual=2`.
- Formal effect-size threshold: `lfc_test=1` (absolute $\log_2\text{FC} > 1$ corresponds to >2 -fold).
- Script 01 ends with `date/time` and `sessionInfo()`. Script 02 does the same.
- Script 02 exports the master ranking, candidate evidence, rescued candidates, fragile original hits, leave-one-out summary, weight-sensitivity summary, and a GSEA-ranked-gene table.

11. Compact methodology map

Challenge	Methods	What failure would mean
Data quality	PCA, correlations, distances, Cook's, count plots, low-count sensitivity	Candidate/sample may be unstable before formal ranking.
Effect estimate	Shrinkage, <code>lfcSE</code> , effect-size test, threshold grid	Large raw effect may be uncertain or threshold-dependent.
Sample robustness	Leave-one-out	One replicate may be essential to the conclusion.
Model robustness	<code>edgeR</code>	Conclusion may depend strongly on DESeq2 implementation.
Technology robustness	limma microarray + agreement metrics	RNA-seq signal may not reproduce on another platform.
Decision robustness	rank product, percentiles, Pareto, 5,000 weights	Candidate may only look strong under a particular scoring preference.
Biological usefulness	GO/STRING/literature/annotation	Statistically robust gene may still lack a compelling or testable biological hypothesis.

12. Suggested wording for a Methods section

RNA-seq differential-expression analyses were performed from the supplied gene-count matrix using DESeq2 with a ~condition design and normoxia as the reference level. In addition to the practical workflow, sample quality was evaluated using variance-stabilized PCA, Pearson and Spearman sample correlations, sample-to-sample distances, Cook's-distance diagnostics and candidate-level normalized-count plots. Sensitivity to low expression was assessed by repeating DESeq2 after retaining genes with at least 10 counts in at least three samples.

For candidate prioritization, DESeq2 $\log_2$ fold changes were shrunken using `apeglm` when available, and a formal effect-size test evaluated evidence for absolute $\log_2$ fold changes greater than 1. Sensitivity to FDR and effect-size cutoffs was evaluated over a grid of thresholds. Robustness to individual biological replicates was assessed by leave-

one-sample-out refitting of DESeq2. An independent RNA-seq analysis was performed with edgeR using filterByExpr, TMM normalization, dispersion estimation and quasi-likelihood testing.

Microarray log(H/N) ratios were reanalyzed with limma using standard empirical Bayes, robust empirical Bayes and TREAT effect-size testing. RNA-seq and microarray effects were integrated by gene and assessed for direction, magnitude and cross-platform difference. Candidate evidence was subsequently summarized using rank-product consensus and percentile-normalized statistical scores. Non-dominated candidates were identified using a Pareto-front analysis across effect, significance, precision, microarray significance and platform agreement. Finally, ranking sensitivity was evaluated over 5,000 random positive weighting schemes, and sample- and decision-robustness were combined using the geometric mean of leave-one-out FDR stability and top-five weight-selection frequency.

13. Source basis for this report

This document is based on the actual extended R Markdown workflows generated and run in this project:

01_rnaseq_preprocessing_extended.Rmd / HTML and 02_integrated_DE_and_candidate_ranking_v3.Rmd / HTML.

The optional BAM/counting commands are described as planned sensitivity analyses because the script marks them eval=FALSE.